ECE 329 **Homework 7** **Due: Mon, Nov 10, 2025, 11:59PM**

7 problems totaling 70pts.

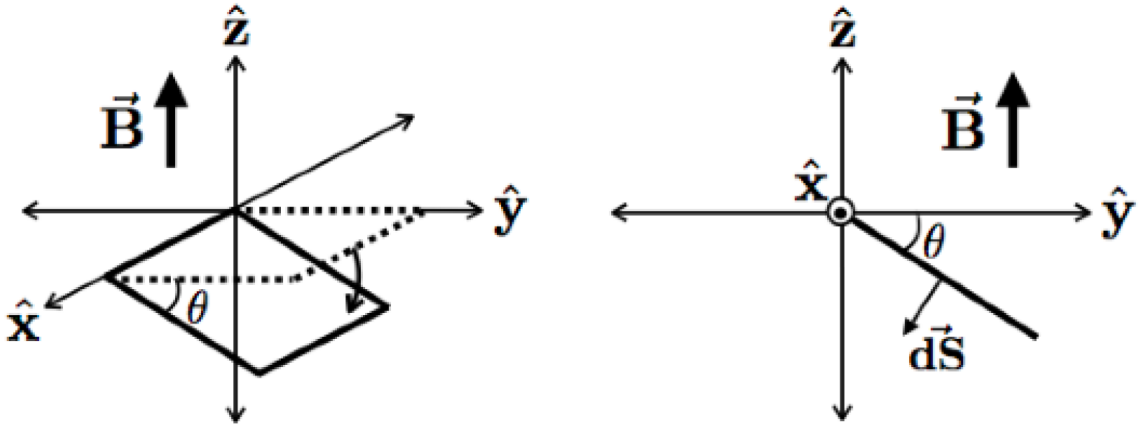
1. (10pts) Given the time-varying magnetic field $\mathbf{B} = B_0(t\sin(\omega t)\ \hat{x} + \cos(\omega t)\ \hat{z})$ Wb/m^2, find the emf $\mathcal{E}$ for the following closed paths C:

   a) (3pts) C is a rectangular path going from $(0,0,0)$ to $(1,0,0)$ to $(1,1,0)$ to $(0,1,0)$ back to $(0,0,0)$, with distance along the path measured in units of meters.

   b) (3pts) C is a rectangular path having the same coordinates as defined above, but with the path direction reversed.

   c) (4pts) C is a triangular path going from $(0,0,0)$ to $(1,0,0)$ to $(1,1,0)$ back to $(0,0,0)$, with distance along the path measured in units of meters.

2. (10pts) A square loop of wire of some finite resistance R and 1 m^2 surface area is located within a region of constant magnetic field $\mathbf{B} = 2\hat{z}$ $\mathrm{Wb/m}^2$ as illustrated in the following diagrams (perspective and side views are shown).



a) (2.5pts) What is the magnetic flux Ψ through the loop when the orientation angle of the loop is $\theta = 0^o$? In your flux calculation make use of $d\mathbf{S}$ orientation shown in the diagram on the right.

b) (2.5pts) What is the flux Ψ as a function of angle θ (using the same sign convention as in part a)?

c) (2.5pts) Assuming that angle θ is time varying at a rate of $\frac{d\theta}{dt} = \pi \frac{\mathrm{rad}}{\mathrm{s}}$, what is the emf $\mathcal{E}$ around the loop at the instant when $\theta = 45^o$?

d) (2.5pts) In what direction will a positive induced current flow around the loop at the same instant? You may indicate your answer by drawing labeled arrows around the loop graphically. Justify your answer.

3. (10pts) A conducting wire loop of radius $r = 1$ m is moved with velocity $\mathbf{v} = 4\hat{x}$ m/s in a region where the background magnetostatic field is described by

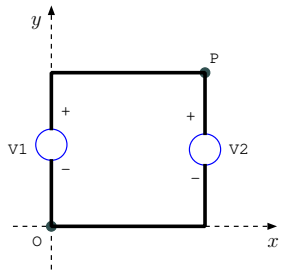
$$\mathbf{B}(x, y, z) = \hat{z}\, 25 \times 10^{-6}(1 + x/L)\,\mathrm{T},$$

where $L = 1000$ m. The center of the loop coincides with the origin $(x, y, z) = (0, 0, 0)$ at $t = 0$ and the plane of the loop coincides with $z = 0$ plane.

a) (5pts) Obtain an expression for the induced emf $\mathcal{E}(t)$ of the loop in motion for $t > 0$. Since $r \ll L$, the magnetic field across the loop can be considered nearly constant at each instant in time.

b) (5pts) What is the magnitude of the loop current if the loop resistance is $4\,\Omega$?

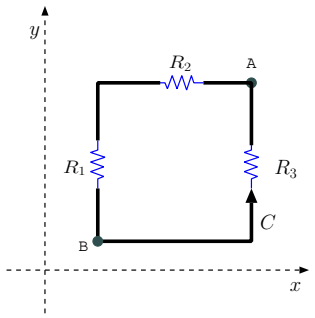
Interesting facts: The strength of Earth's magnetic field is just about 25×10^{-6} T at equatorial latitudes. However, the scale length L associated with the spatial variation of Earth's magnetic field is much longer than 1000 m.

4. 

(10pts) Voltmeters V1 and V2 are connected between points O=$(0,0)$ and P=$(1,1)$ m on the xy-plane as shown in the figure on the left along the sides of a 1x1 m square. The magnetic flux linking the same square area S computed with $d\mathbf{S} = \hat{z}dxdy$ is given as $\Psi(t) = \int_S \mathbf{B} \cdot d\mathbf{S} = -6t$ Webers. If at $t = 0$, voltmeter V1 displays a voltage of $V_1 = 2$ V, what voltage value V_2 is displayed by voltmeter V2 at the same time instant? Carefully explain your reasoning.

Hints: (a) note that $V_1 - V_2 \neq 0$ but ..., and (b) each voltmeter by design reports the numerical value of the line integral $\int_{P+}^{P-} \mathbf{E} \cdot d\mathbf{l}$ taken along the path constituted by the leads of the voltmeter from its (+) probe tip at $P+$ to its (-) probe tip at $P-$.

5.



(10pts) Consider the circuit depicted on the left. Resistors $R_1 = 1\,\Omega$, $R_2 = 3\,\Omega$, and $R_3 = 2\,\Omega$ constitute a square loop of 1 m^2 physical area in a region where there is a uniform magnetic field $\mathbf{B} = B_o\hat{z}$. It is known that $dB_o/dt = -12$ T/s. Assume there is no magnetic field outside the squre loop. Answer the following questions:

a) (2pts) Compute the induced emf $\mathcal{E}$ in the circuit in the circulation direction C indicated on the diagram.

b) (2pts) Compute the loop current I induced in the circuit in the circulation direction C indicated on the diagram.

c) (4pts) If a voltmeter is connected *diagonally* between points A and B with the (+) lead contacting point A (and in effect splitting the square loop into a pair of equal area triangles) what will be the voltmeter reading?

d) (2pts) Describe how you would re-connect the voltmeter between points A and B to directly read the voltage drop of R_3 on the right.

6. (10pts) If

$$\mathbf{E} = -\cos(\omega t + \beta x)\,\hat{y}\,\frac{\mathrm{V}}{\mathrm{m}},$$

$\frac{\omega}{\beta} = c$, and $\mu = \mu_o$, find the corresponding $\mathbf{H}$ by using Faraday's law

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}.$$

Hint: find a suitable time varying anti-derivative for $\nabla \times \mathbf{E}$.

7. (10pts) If

$$\mathbf{H} = \sin(\omega t - \beta z)\,\hat{x}\,\frac{\mathrm{A}}{\mathrm{m}},$$

$\sigma = 0$, $\frac{\omega}{\beta} = \frac{2}{3}c$ and $\epsilon = 2.25\epsilon_o$, find the corresponding $\mathbf{E}$ by using Maxwell-Ampere's law

$$\nabla \times \mathbf{H} = \mathbf{J} + \frac{\partial \mathbf{D}}{\partial t}$$

in which $\mathbf{J} = \sigma\mathbf{E}$ and $\mathbf{D} = \epsilon\mathbf{E}$.